

# Re-analysis of GEO GSE54825: transcriptional response of engineered *Saccharomyces cerevisiae* to cellobiose versus glucose

A reproducible pipeline with a quantitative comparison to the original publication

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## 1 Background

*Saccharomyces cerevisiae* cannot metabolise cellobiose. Engineering a heterologous cellobiose utilisation pathway — a cellodextrin transporter (*cdt-1*) and an intracellular  $\beta$ -glucosidase (*gh1-1*) from *Neurospora crassa* — enables growth on this sugar, but fermentation remains slower than on glucose [1]. The original study of the data set re-analysed here (GEO GSE54825; [2]) profiled the transcriptome of the engineered strain on cellobiose versus glucose under anaerobic conditions and reported 519 differentially expressed genes, mitochondrial activation, reduced amino-acid and thiamine biosynthesis, and two engineering leads (*SUT1* overexpression, *HAP4* deletion).

That study was analysed with a commercial platform (CLC Genomics Workbench 6.5), a private reference assembly, and a statistical framework (Baggerley's test) that is no longer standard. This report re-analyses the same raw reads with a modern, fully scripted pipeline and asks a question the original could not answer: how much of the original result is reproducible, and what is not assessable at all?

## 2 Research question

What is the global transcriptional difference between cellobiose- and glucose-grown engineered *S. cerevisiae* under anaerobic exponential growth; which biological processes are involved; and which observations from the original study can be quantitatively reproduced with an independent pipeline?

## 3 Dataset

The seven input files (paper and supplements of both studies) are archived read-only with md5 checksums; the sequencing data are public and were verified against the md5 values published by ENA.

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## 4 Experimental design

Two conditions × three biological replicates, a single batch, anaerobic exponential phase (80 g/L starting sugar, harvested at ≈50 g/L residual sugar). The original study used an unpaired two-group comparison; replicate numbering (*JCYL001B/001D*, ...) hints at pairing but this cannot be verified from public metadata. The primary model is therefore unpaired (*~ condition*) and a paired model (*~ replicate + condition*) is run as a sensitivity analysis (see the Differential expression section). No spike-in controls and no independent batch are available.

## 5 Methods

| Step     | Tool, version and settings                                                 |
|----------|----------------------------------------------------------------------------|
| Raw data | ENA FTP download; md5 verified against ENA-published values                |
| Read QC  | FastQC 0.12.1; fastp 1.3.6 (light trimming, length >= 36 bp); MultiQC 1.35 |

| Step                    | Tool, version and settings                                                                                                                                                |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reference (host)        | RefSeq GCF_000146045.2 (S288C R64-1-1, 16 chromosomes + mitochondrion); transcripts extracted with gffread 0.12.9; annotation from the SGD GAF (release R64-5-1)          |
| Reference (trans-genes) | Host-vector-insert sequences adjudicated from the reads themselves (see the Quantification section)                                                                       |
| Quantification          | Salmon 2.7.0, decoy-aware index (k = 31; k = 25 as sensitivity), <code>-libType A -gcBias -seqBias</code> , fragment-length prior set to the observed read length (49 bp) |
| Gene-level matrix       | tximport 1.34.0, <code>countsFromAbundance = lengthScaledTPM</code>                                                                                                       |
| Differential expression | DESeq2 1.46.0; <code>primary ~ condition</code> , <code>sensitivity ~ replicate + condition</code> ; LFC shrinkage (apeglm) for display only                              |
| Multiple testing        | Benjamini-Hochberg                                                                                                                                                        |
| Functional enrichment   | clusterProfiler 4.14.0 (ORA + GSEA) with self-built TERM2GENE tables; KEGG (sce) cached                                                                                   |
| Report                  | Quarto 1.9.38 (HTML and PDF via Typst)                                                                                                                                    |

Thresholds. Primary DEGs are BH-adjusted  $p < 0.05$  and  $|\log_2\text{FC}| > 1$ . A relaxed ( $p < 0.1$ ) and a stringent tier ( $p < 0.01$  and  $|\log_2\text{FC}| > 1.5$ ) are also reported, plus a tier matched to the original study ( $\text{FDR} \leq 0.001$  and  $|\log_2\text{FC}| \geq 1$ ). All thresholding uses unshrunk (MLE)  $\log_2\text{FC}$  values; apeglm-shrunk values are used only for ranking and plots. The enrichment universe is the set of expressed genes (count  $\geq 10$  in at least three samples): [r](#) [r](#) 6049 genes.

## 6 Quality control

The 2014 data are single-end 50 bp reads from a Genome Analyzer II. FastQC reported Sanger/Illumina 1.9 encoding, so no Phred64 handling was required. Light trimming retained 95.1% of 238.6 million reads and raised the Q30 fraction from 0.88–0.93 to 0.93–0.96; adapter content is negligible ( $\approx 150$  reads per sample after trimming, 0.01%).

Read fate. Reads were attributed in three tiers: (i) end-to-end alignment to the host genome plus transgenes (94.5–95.5%), (ii) local-mode re-alignment of the remainder (recovering 19.9–22.8% of the unmapped fraction, attributable to the strict end-to-end mode and to read errors in these older data), and (iii) a diagnostic reference containing the pRS426 backbone, the native 2  $\mu\text{m}$  plasmid and TruSeq adapters (a further 35.6–40.4% of the residual). A small, quantified fraction (2.2–3.0% of the library) remains unexplained; candidate explanations are listed in the Limitations section. Multi-mapping is 23.0–24.9%, driven by Ty elements and the rDNA repeat, and the rDNA locus is the most deeply covered region of the reference (RDN25-1, mean depth  $23\times$ ).

Plasmid copy number. Reads from the engineered plasmid backbone are  $1.65\times$  more abundant on cellobiose (10,385  $\rightarrow$  17,188 reads per million), whereas reads from the native 2  $\mu$ m plasmid are unchanged (3,358  $\rightarrow$  3,012). This is independent evidence that any copy-number effect is specific to the engineered plasmid.

## 7 Quantification

The host reference and annotation were frozen (RefSeq [GCF\\_000146045.2](#), SGD annotation release R64-5-1, all files md5-recorded). Two properties of this data set required specific handling.

The transgenes are not what the literature suggests. The RNA-seq samples carry pRS426-BT, whose inserts derive from *N. crassa* cDNA. Reads-derived consensus sequences match the public CDS of *cdt-1* (NCU00801) and *gh1-1* (NCU00130) base-for-base (0 differences). The codon-optimised [gh1-1a](#) published in the supplement (Figure S8) is absent from the samples (2 reads in 6 million) and differs from the native CDS at 25% of positions: using it as a reference would have silently destroyed transgene quantification. The C-terminal tag is a superfolder GFP: the consensus differs from the humanised EGFP CDS at only 7 nucleotides, and its translation carries all six superfolder substitutions (S30R/Y39N/N105T/Y145F/I171V/A206V) on an EGFP backbone. The *cdt-1*-tag junction was read directly from soft-clipped reads: ...GTTGCT + ATCGAT (Clal) + GGTAGTGGTAGT (Gly-Ser-Gly-Ser) + GTGAGC..., consistent with the cloning strategy described in the earlier work [1]. Because part of the transgene reference is derived from the sample reads, transgene results are reported as ratios only.

Mitochondrially encoded genes must be measured, not assumed. The RefSeq RNA FASTA contains no mitochondrial protein-coding mRNA, so transcripts were extracted from the GFF with gffread (6480 host transcripts, including 19 mitochondrial mRNAs) plus the two transgenes.

Salmon with a decoy-aware index mapped 80.0–83.5% of reads ( $k = 31$ ); library type was inferred as unstranded (U) in all twelve runs, consistent with TruSeq v1 chemistry. Quantification was validated in three independent ways: (i) TPM versus the original RPKM correlates with Spearman = 0.880 over 5796 genes; (ii) adding or removing the genome decoy changes TPMs by a median of 0.027 log2 units (17 of 5960 genes exceed 1); and (iii) two calibration genes reproduce the original values almost exactly (URA3 ratio 1.364 versus 1.354; PGK1 ratio 0.749 versus 0.745).

Reported **EffectiveLength** values are anomalous in Salmon 2.7.0 (Rust port): host transcripts receive a median effective length of  $0.465 \times$  length, and the transgene entries receive values above  $10 \times$  length. This does not affect the conclusions above; the checks in (i)–(iii) and the decoy comparison were performed specifically to bound the risk, and all transgene statements are based on ratios.

## 8 Sample-level analysis

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 !Fig. 1 PCA of VST-normalised expression. Conditions separate on PC1 (93.4% of
```

```

variance); within-condition replicates are tightly clustered (Spearman rho 0.994-0.998).]
(figures/pca.png){#fig-pca fig-alt='PCA plot of six samples showing separation by carbon
source'}
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!["**Fig. 2** Sample-to-sample Spearman correlation of VST-normalised expression.](figures/
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Conditions separate on PC1 (93.4% of variance) and replicates within a condition are
extremely
concordant (Spearman  $\rho$  0.994–0.998) compared with the between-condition range (0.886–
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supporting H0.

# Differential expression {#sec-de}

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  [~condition FDR<=0.001 & |LFC|>=1], [6050], [ 632], [ 341], [ 291], [2],
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!["**Fig. 3** Volcano plot of the primary model (`~ condition`). Thresholds: BH-adjusted p
< 0.05 and |log2FC| > 1; fold changes are unshrunk (MLE) values, which are the values
used for all thresholding.](figures/volcano.png){#fig-volcano}
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!["**Fig. 4** Row-scaled VST expression of the 50 most significant DEGs.](figures/
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!["**Fig. 5** p-value distribution of the primary model (no systematic anti-
conservatism; Cook's cutoff exceeded by 1 of 6049 genes.)(figures/model_diagnostics/
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!["**Fig. 6** MA plot with apegglm-shrunken log2FC (display only; thresholding uses MLE values).](figures/model_diagnostics/ma_plot.png){#fig-ma}
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:::

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The primary model yields **644 DEGs** (346 up, 298 down); the tier matched to the original thresholds yields **632** (341 up, 291 down). The paired sensitivity model yields 643 DEGs and agrees with the primary model almost perfectly (Jaccard 0.986; 100% directional concordance; Pearson correlation of log2FC = 1.0000), so the conclusions do not depend on the pairing assumption. Model diagnostics show no systematic anti-conservatism and only one gene above the Cook's distance cutoff.

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[50/316], [1.65e-28], [1.62e-26], [ 50],
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[8.87e-20], [ 32],

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!["**Fig. 7** GO slim over-representation analysis (top 10 terms per gene set; BH-adjusted p).](figures/enrichment/go_slim_ora.png){#fig-goslim}
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[5.53e-16], [1.47e-13],
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:::
:::

```

The original study used FunSpec with Bonferroni correction (p 0.01); this re-analysis uses clusterProfiler with Benjamini-Hochberg correction and a self-built GO slim mapping (SGD `go\_slim\_mapping.tab`) as well as full GO annotations propagated to ancestor terms via the true

path rule. The direction and identity of the enriched processes reproduce the original conclusions: upregulated genes are dominated by the respiratory chain, oxidative phosphorylation, ATP synthesis, mitochondrial and TCA-cycle terms, whereas downregulated genes

are dominated by amino-acid biosynthesis (methionine, cysteine, arginine, histidine) and **thiamine (vitamin B1) biosynthesis**, with the same signal in KEGG (oxidative phosphorylation, biosynthesis of amino acids, cysteine/methionine and histidine metabolism).

One observation not reported originally: the strongest negative GSEA signal involves Ty retrotransposon-related terms. Because these sequences are heavily multi-mapping, this finding is reported with caution and is not used for any biological claim.

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# Biological interpretation {#sec-interpretation}
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[NA], [replicated (differentially expressed)],
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  [salmon_TPM], [gh1-1], [ 26.4], [ 84.4], [3.190], [TPM (长度归一化)],
  [URA3_proxy], [URA3(YEL021W)], [ 432.0], [ 589.0], [1.360], [质粒拷贝数代理(原文锚定 1.354)],
  [PGK1_control], [PGK1(YCR012W)], [NA], [NA], [0.749], [同启动子反证(原文锚定 0.745)],
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[reads_1M_subsample], [gh1-1], [1200.0], [2680.0], [2.220], [裁决实验 bowtie2 读数 (PLAN
3.4.3) ],
)

```

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!["**Fig. 10** Independent estimators of the transgene fold change (cellobiose/glucose) and
of the plasmid copy-number proxy URA3. Read-based estimators (1.89-2.01 for cdt-1EGFP,
2.22-2.32 for gh1-1) bracket the values reported in the original study (1.87 and 2.16).]
(figures/transgene_anchor.png){#fig-anchor}
:::
:::

```

The nine pre-registered hypotheses were scored with set-level statistics (Wilcoxon signed-rank against zero, Fisher's exact test for differential-expression enrichment, and the number of DEGs inside each set). Every hypothesis is supported:

- **H1 (mitochondrial activation)** replicates across five independent gene sets (respiratory chain complex +1.91, ATP synthesis +1.79 with 15 of 16 genes differentially expressed, oxidative phosphorylation +1.79, electron transport chain +1.55, TCA cycle +0.86). Mitochondrially encoded genes are excluded from this test (see *Limitations*).
- **H2 (amino-acid and thiamine biosynthesis down)** replicates in all six subsets, including thiamine biosynthesis (-1.22, 7 of 16 genes differentially expressed, Fisher  $p = 2.7 \times 10^{-3}$ ).
- **H3 (glucose signalling only partially activated)** is supported in the sense the original reported: the PKA set shifts only slightly (+0.45) and the Rgt1 and Snf1-Mig1 sets show no systematic shift.
- **H4 (hexose-transporter rearrangement)** has no net directional shift (mean log2FC -0.14) but six of eleven transporters are differentially expressed: a family-level rearrangement rather

```

    than a uniform response.
- H5-H8 replicate, including the engineering leads: SUT1 +1.77, HAP4 +2.61 and DAL80
+1.22.
- H9 (growth-rate/glucose-repression confounder) – the ribosomal-protein gene set
(132 genes)
    shifts by +0.34 ( $p = 1.4 \times 10^{-21}$ ). A confounder therefore exists and is measurable; its
    direction suggests relief of glucose repression rather than a slow-growth effect, because
    ribosomal-protein genes are typically *up* at higher growth rates in yeast.

# Comparison with the original study {#sec-comparison}

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  [overlap], [ 496.000],
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  [background_genes], [5780.000],
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!["Fig. 8" Per-gene log2FC agreement with the original study (Spearman = 0.973 over 5771
genes).](figures/fc_agreement.png){#fig-fc}
:::
:::

::: {.cell}
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!["Fig. 9" Precision/recall of the original 519 DEGs as a function of threshold.](figures/
threshold_sweep.png){#fig-sweep}

```

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transcripts inefficiently; 5-18 chrM reads per 1e6; 0/19 mt genes reach 1x],
  [absolute transgene abundance], [the tag/junction part of the transgene reference is
derived from the sample reads themselves (circularity)],
  [absolute RPKM/TPM matching the original], [CLC multi-mapping and 'By totals' RPKM are
not reproducible outside CLC; only gene-level trends are comparable],
  [cross-tool mapping statistics], [original 92.3/83.4/76.3% were produced by CLC's internal
algorithms; only order-of-magnitude comparison is meaningful],
  [transgene C-terminal His6 cleavage state], [requires protein-level data, out of scope],
)

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  [dataset_s1_transgenes], [ 2], [外源条目数 (cdt-1EGFP, gh1-1)], [ok],
  [dataset_s2_deg_rows], [ 519], [原文报告的 519 个 DEG], [ok],
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  [dataset_s3_down_genes], [ 256], [Dataset S3 区块标题声明的 DOWN 基因数], [ok],
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  [check_sum_of_parts], [ 503], [对账: UP + DOWN + 两侧 not found], [ok],
  [check_gap_vs_519], [ 16], [与 519 的差额; 需在报告 5.12 解释 (候选: 外源条目 gh1-1 无 GO 注

```

```
释) ], [ok],  
)
```

```
:::  
:::
```

Agreement is high at every level that can be compared:

```
| Level | Result |  
| --- | --- |  
| Per-gene log2FC (5771 genes) | Spearman **0.973**, Pearson 0.840 |  
| DEG overlap (632 vs the original 519) | **496 shared**, Jaccard 0.757, hypergeometric p  
≈ 0 |  
| Direction concordance in the overlap | **100%** |  
| Count reconciliation | original 519 = 244 up + 256 down + 18 unmappable + *gh1-1* (no GO  
annotation) |  
| Transgene fold changes (ratio-based) | 1.89–2.01 (cdt-1EGFP) and 2.22–2.32 (gh1-1) versus  
1.87 and 2.16 originally |  
| Plasmid copy-number proxy (URA3) | 1.364 versus 1.354 originally |
```

Absolute RPKM/TPM values cannot match the original: CLC's internal multi-mapping handling is not reproducible outside CLC (only 76.3% of reads were uniquely mapped there), and percentages such as 92.3/83.4/76.3% are software-specific quantities. What is reproducible is the gene-level trend, and the trend is reproduced almost exactly.

# Limitations {#sec-limitations}

- **\*\*Enrichment framework differs\*\*** from the original (clusterProfiler + BH versus FunSpec + Bonferroni), so enrichment p-values are not directly comparable; only the identity and direction of the enriched processes are.
- **\*\*Mitochondrially encoded genes are not assessable\*\***: poly-A selection captures them poorly (5–18 chrM reads per million; none of the 19 mitochondrial genes reaches 1× coverage), which is expected because budding-yeast mitochondrial mRNAs lack the long poly(A) tails used for capture [Chang2012]. H1 is therefore tested on nuclear-encoded mitochondrial genes only.
- **\*\*Transgene abundance is circular to a degree\*\***: the tag and junction sequences are derived from the sample reads. Ratios are robust to this (the same reference bias applies to both conditions), absolute TPM is not.
- **\*\*Plasmid copy number varies between conditions\*\*** (URA3 1.364×; backbone reads 1.65×).

Dividing

the transgene ratio by the proxy (1.47 for cdt-1EGFP, 1.70 for gh1-1) lowers the effect but does

not remove it, so part of the transgene increase reflects copy number rather than transcription.

- **Terminator/promoter crosstalk** from the plasmid affects PGK1 and CYC1 minimally but cannot be

fully separated; CYC1 is itself a mitochondrial-activation gene and changes by 2.5× in the

original data.

- **Unmapped reads are only partly explained**: 2.2–3.0% of the library remains unattributed;

candidates include read errors in GA II data, non-yeast contamination, regions that are collapsed or absent in the R64 assembly (subtelomeric/Y' elements), and chimeric reads spanning

junctions that exist in no reference (promoter–CDS, CDS–terminator).

- **No spike-in, no independent batch, n = 3** limit absolute quantification and variance estimation; surrogate-variable methods are not applicable with six samples.

- **EffectiveLength** semantics are anomalous in Salmon 2.7.0 (see the *Quantification* section); the

conclusions were shown to be insensitive to it, but absolute TPM values should not be over-interpreted.

- **Pairing could not be verified**; the paired model is reported as a sensitivity analysis only.

# Engineering implications {#sec-engineering}

The re-analysis reproduces the two engineering leads of the original study and places them in a

quantified context: SUT1 (+1.77) and HAP4 (+2.61) are among the most strongly induced transcription

factors, and DAL80 (+1.22) also qualifies. Because HAP4 is *induced* on cellobiose while its deletion improves fermentation, its regulatory role is non-monotonic – a caution against reading

transcriptional direction as a direct engineering instruction. The yeast mitochondrial and amino-acid/thiamine pathways together account for the bulk of the response, which suggests that

engineering strategies aimed at cofactor and precursor supply (rather than at sugar transport

alone) are the most promising.

# Reproducibility {#sec-reproducibility}

Everything in this report is generated by numbered scripts in `scripts/`:

```bash

```
conda env create -f environment.yml && conda activate yeast-cellobiose-rnaseq
bash run_all.sh          # idempotent; skips completed steps
```

| Artefact                                               | Location                                                                                   |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Raw-data checksums (self-computed and ENA-published)   | <a href="#">checksums/</a>                                                                 |
| Reference and annotation versions                      | <a href="#">refs/VERSIONS.md</a>                                                           |
| Provenance of every non-public sequence                | <a href="#">refs/custom/PROVENANCE.md</a>                                                  |
| Sequence adjudication and junction evidence            | <a href="#">refs/custom/</a><br><a href="#">{sequence_adjudication.tsv, junctions.tsv}</a> |
| Environment lock (conda-explicit + R package snapshot) | <a href="#">env/</a>                                                                       |
| Parameter and scope decisions                          | <a href="#">docs/decisions/</a>                                                            |
| Stage acceptance records                               | <a href="#">docs/acceptance/</a>                                                           |

The R environment is locked with conda rather than renv because renv project activation masks the conda library in this setup ([docs/decisions/r\\_environment.md](#)).

## 9 Conclusion

Re-analysing GSE54825 with an independent, fully scripted pipeline reproduces the original study almost completely at the level at which comparison is meaningful: 97.3% rank agreement of per-gene fold changes, 496 of the original 519 DEGs recovered, 100% directional concordance, and the same enriched processes. Three results go beyond the original: the transgene reference sequences were reconstructed and validated from the data (revealing that the published codon-optimised gh1-1a is not the allele in these samples), the plasmid copy-number contribution to the transgene signal was quantified, and the growth-rate/glucose-repression confounder was measured rather than assumed. The limitations that cannot be removed — mitochondrially encoded genes, absolute transgene abundance, and CLC-specific mapping statistics — are stated explicitly rather than papered over.

## References

### Bibliography

- [1] J. M. Galazka, C. Tian, W. T. Beeson, B. Martinez, N. L. Glass, and J. H. D. Cate, “Cellodextrin transport in yeast for improved biofuel production,” *Science*, vol. 330, no. 6000, pp. 84–86, 2010, doi: 10.1126/science.1192838.
- [2] Y. Lin et al., “Leveraging transcription factors to speed cellobiose fermentation by *Saccharomyces cerevisiae*,” *Biotechnology for Biofuels*, vol. 7, p. 126, 2014, doi: 10.1186/s13068-014-0126-6.